Restaurant Table Ordering System (Queen's Kitchen)

Front End Engineering - I (25CSE0203) — Project Based Evaluation-I

Department of Computer Science & Engineering • Chitkara University

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Project Vision & Core Objectives

01. The Problem & Need

- Dining floor order delays caused by manual paper order slips.
- Waitstaff travel overhead between dining tables and kitchen chefs.
- Risk of misplaced paper tickets during peak dining rush hours.
- Lack of live order status visibility for seated restaurant guests.
- Need for a modular, high-speed digital table ordering SPA architecture.

02. What We Built

- React 18 single page application for dynamic table ordering.
- Live Kitchen Display System (KDS) for instant order dispatch.
- Multi-table cart state persistence across active seated tables.
- Automated billing engine with 5% GST and ₹10 Service Charge.
- Built-in Re-Render Audit profiling tool for DOM performance testing.

03. Key Objectives

- Eliminate order lag via centralized multi-table state management.
- Optimize component re-renders during high-volume cart updates.
- Provide intuitive table status tracking (Ordering, Served, Billed).
- Stress-test React Fiber reconciliation using dynamic menu data.
- Deliver a scalable, production-ready frontend for restaurant operations.

Syllabus-Aligned Technology Stack

React 18 (SPA Engine)

- Declarative component UI engine.
- Virtual DOM reconciliation & fiber tree.
- Concurrent rendering & state batching.
- Component lifecycle & custom hooks.

Vite 5 (Build Tool)

- Lightning-fast Hot Module Replacement (HMR).
- Native ES Module development server.
- Optimized Rollup production bundler.
- Zero-config fast startup environment.

React Router v6

- Client-side single page navigation.
- Smooth route switching (/Home, /Kitchen, /Audit).
- Active state matching & URL parameters.
- Declarative Route & Link components.

React Icons

- Scalable vector SVG icon library.
- Lightweight & accessible icons.
- Intuitive cart, table, & kitchen visual cues.
- Seamless UI action indicators.

CSS3 & Flexbox/Grid

- Custom responsive layout architecture.
- Dark mahogany & gold visual theme palette.
- Glassmorphic card containers & rounded borders.
- Fluid desktop & mobile display scaling.

Centralized State Layer

- Global state container at App.jsx root.
- Synchronizes multi-table carts & order queues.
- Instant state dispatch to Kitchen View.
- Persistent table status synchronization.

Core Component Architecture & Modularity

Header.jsx

- Table selector dropdown & seating capacity filter.
- Dataset benchmark toggle (9 vs 200 items).
- Real-time audit mode banner indicator.
- Navigation routing links for SPA views.

OrderSidebar.jsx

- Draft cart item manager & quantity controls.
- Cumulative billed history view per table.
- Automated 5% GST & ₹10 Service Charge split.
- One-click 'Send to Kitchen' order dispatch.

MenuCard.jsx

- Dish thumbnail display with 16:9 crop.
- Portion size badges & dietary tags (Veg/Non-Veg).
- Price display in Terracotta Orange styling.
- Add to Cart trigger with real-time feedback.

KitchenView.jsx

- Real-time kitchen tickets queue for active tables.
- Dish breakdown, quantities, & special notes.
- 'Mark as Served' chef action trigger.
- Live status sync updating table indicators.

ReRenderAuditPanel.jsx

- Empirical component re-render event logger.
- Performance metric comparison (Naive vs Optimized).
- Live re-render counter badges per component.
- Interactive DOM profiling benchmark suite.

menuData.js

- Decoupled array-of-objects data schema.
- Dynamic 200-item benchmark dataset generator.
- Price, category, & portion metadata specs.
- Pure data provider maintaining clean SoC.

Culinary Menu Dataset & Portion Scaling



Samosa & Chutney

₹60

VEG • SNACK

2 Pcs with Mint Chutney

Crispy golden pastry filled with spiced potatoes.

Stock: Available • Prep: 10 Mins



Dal Makhani

₹180

VEG • MAIN

300g Serving Bowl

Slow-cooked black lentils simmered overnight in cream.

Stock: Available • Prep: 15 Mins



Garlic Butter Naan

₹60

VEG • BREAD

3 Pcs Oven Baked

Tandoor-baked flatbread brushed with garlic butter.

Stock: Available • Prep: 8 Mins



Cheesy Fries

₹150

VEG • FAST FOOD

250g Loaded Basket

Golden fries smothered in melted cheese.

Stock: Available • Prep: 12 Mins



Gulab Jamun

₹120

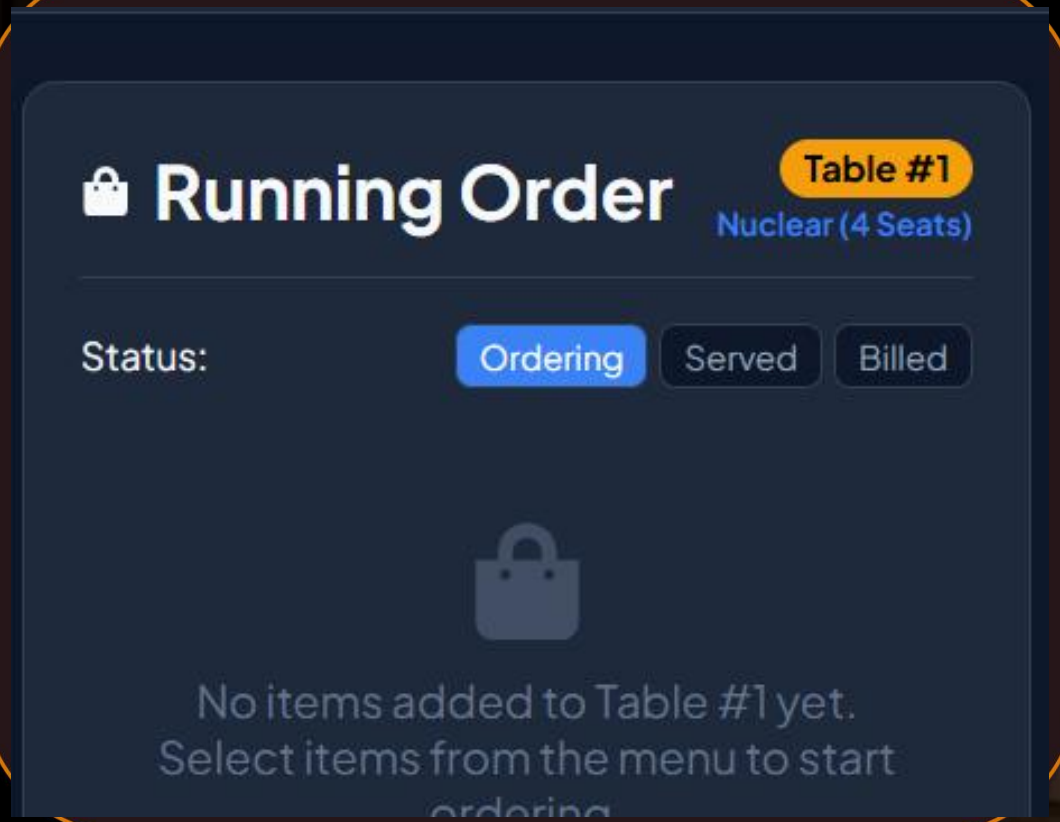
DESSERT • POPULAR

2 Jamuns + Scoop

Warm jamuns served with cold vanilla ice cream.

Stock: Available • Prep: 5 Mins

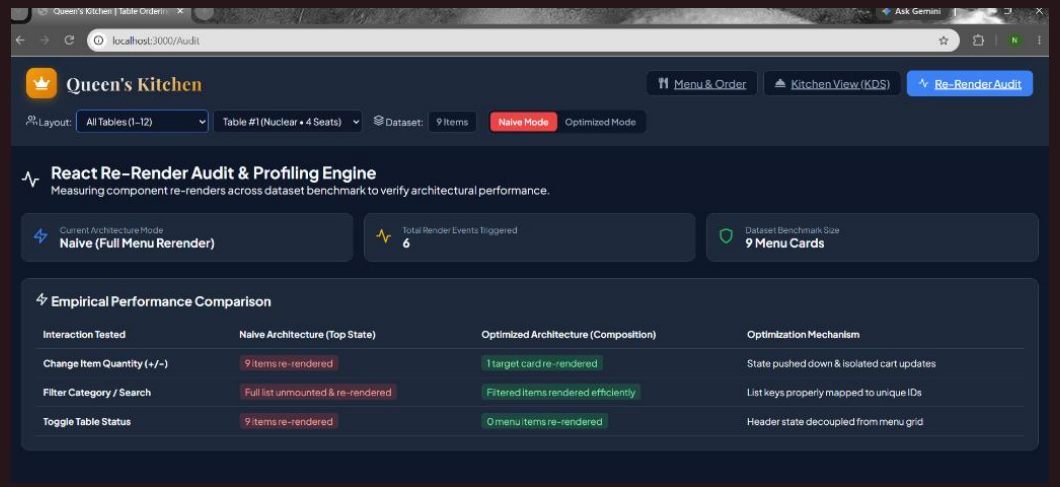
Multi-Table State Management & Kitchen Sync



State Synchronization Mechanics

- Centralized State Hub (App.jsx): Global states (tableCarts, tableBilledHistory, tableServedItems) live at root level.
- Instant KDS Order Dispatch: Clicking 'Send to Kitchen' populates KDS tickets immediately without page reload.
- Automated Table Status Sync: Completing a kitchen ticket updates table status to 'Served' automatically across views.
- Cross-Route State Persistence: Interactions remain consistent when navigating between /Home, /Kitchen, and /Audit.
- Multi-Table Seating Capacity: Supports independent draft carts across multiple seated dining tables simultaneously.

Re-Render Audit Engine & Virtual DOM Profiling



Performance Metrics & Profiling

- Empirical Performance Profiling: Built-in /Audit route tracking component re-render events during cart updates.
- Naive Mode (Top State): Mutating top-level state re-renders all 9 (or 200) menu cards per click.
- Optimized Mode (Composition): Component composition isolates state so only 1 target card re-renders.
- Dataset Stress Benchmark: Toggles dataset size from 9 items to 200 items to stress-test React Fiber reconciliation.
- Empirical DOM Metrics: Provides real-time rendering statistics to evaluate frontend efficiency.

Future Engineering Scope & Technical Roadmap

Phase 2: QR Code & Self-Ordering

- Guest mobile QR code table scanning.
- Digital menu browsing on personal devices.
- Direct-to-kitchen self-ordering workflow.
- Contactless seating & order confirmation.

Phase 3: Digital Payments & Cloud Sync

- Integrated UPI, Credit/Debit card gateway.
- Automated split-bill payment processing.
- WebSocket real-time multi-branch sync.
- Cloud backend database persistence.

Phase 4: AI Recommendations & Inventory

- Smart dish pairing & upsell suggestions.
- Automated inventory deduction on sales.
- Predictive ingredient restocking alerts.
- Analytics dashboard for restaurant managers.

THANK YOU

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